DSAA3073: Theories in Data Science

Week 4 Validation Sheet: Bias-Variance Tradeoff & Regularization

Total Points: 130

Instructions:
- Write your answers clearly and legibly.
- Show all working for full credit.
- State any assumptions you make.
- For proofs, justify each step rigorously.
- This is a practice validation sheet. AI tools and references are allowed.
- Use this to assess your understanding before the actual test.

Part A: Multiple Choice (40 points)

Each question is worth 5 points. Select the single best answer.

Question 1. [**5 marks**] In the bias-variance decomposition $\text{MSE}(x) = \text{Bias}^2 + \text{Variance} + \sigma^2$, which component represents systematic error due to model assumptions?

A) Variance

B) Irreducible error σ^2

C) Bias

D) The sum of bias and variance

Question 2. [**5 marks**] You fit two models to the same dataset:

- Model A: Training MSE = 0.05, Test MSE = 0.06
- Model B: Training MSE = 0.01, Test MSE = 0.20

What can you conclude?

A) Model A has higher variance than Model B.

B) Model B is overfitting the training data.

C) Model A has lower bias than Model B.

D) Both models generalize equally well.

Question 3. [**5 marks**] In ridge regression, what is the effect of increasing the regularization parameter λ?

A) Bias decreases, variance increases

B) Bias increases, variance decreases

C) Both bias and variance increase

D) Both bias and variance decrease

Question 4. [5 marks] According to Hadamard's well-posedness criteria, which property ensures that small perturbations in data lead to small changes in the solution?

A) Existence

B) Uniqueness

C) Stability

D) Consistency

Question 5. [5 marks] Consider the closed-form ridge regression solution $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$. What is the computational advantage of adding λI?

A) It reduces the number of features.

B) It guarantees $(X^\top X + \lambda I)$ is invertible even when $X^\top X$ is singular.

C) It eliminates the need to compute $X^\top X$.

D) It makes the solution sparse (many coefficients become exactly zero).

Question 6. [5 marks] Which of the following models typically has the highest bias?

A) 15th-degree polynomial

B) Linear model

C) 5th-degree polynomial

D) Neural network with 10 hidden layers

Question 7. [5 marks] In the eigenvalue decomposition of ridge regression, if $X^\top X$ has eigenvalue $\lambda_j = 0.1$, what is the shrinkage factor when $\lambda = 1$?

A) $\frac{0.1}{1.1} \approx 0.09$

B) $\frac{1.1}{0.1} = 11$

C) $\frac{1}{1.1} \approx 0.91$

D) $\frac{0.1}{1} = 0.1$

Question 8. [5 marks] What does it mean when we say a learning problem is "ill-posed"?

A) The training data contains errors.

B) The model is too complex for the data.

C) At least one of Hadamard's criteria (existence, uniqueness, stability) is violated.

D) The test error is larger than training error.

Part B: Mixed Format Questions (90 points)

Question 9. [15 marks] Consider the bias-variance decomposition for regression.

(a) [5 marks] Write down the full decomposition of $\mathrm{MSE}(x) = \mathbb{E}\left(\left(y - \hat{f}_n(x)\right)^2\right)$ where $y = f^*(x) + \varepsilon$ with $\mathbb{E}(\varepsilon) = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2$. Define each component.

(b) [5 marks] Explain intuitively why variance vanishes as $n \to \infty$ but bias does not.

(c) [5 marks] For a linear model $\hat{f}(x) = \beta_0 + \beta_1 x$ fit to data from $f^*(x) = \sin(2\pi x)$, which component dominates the MSE? Justify your answer.

Question 10. [13 marks] Prove the bias-variance decomposition from first principles.

(a) [8 marks] Starting from $\mathrm{MSE}(x) = \mathbb{E}\Big(\big(y - \hat{f}_n(x)\big)^2\Big)$ where $y = f^*(x) + \varepsilon$ with $\mathbb{E}(\varepsilon) = 0$, derive the decomposition $\mathrm{MSE}(x) = \mathrm{Bias}^2 + \mathrm{Variance} + \sigma^2$.

(b) [5 marks] In your proof, identify where independence between the training sample and test noise is used. What would happen if this assumption were violated?

Question 11. [12 marks] Consider polynomial regression on n data points.

(a) [6 marks] Explain qualitatively how bias and variance change as polynomial degree d increases from 1 to $n-1$. What happens at $d = n-1$?

(b) [6 marks] Suppose the true function is $f^*(x) = x^2$, noise $\sigma^2 = 1$, and we fit polynomials of degree $d = 1$ and $d = 2$ on $n = 100$ samples. Which model should have lower test MSE? Explain using bias-variance tradeoff.

Question 12. [15 marks] Derive the closed-form solution for ridge regression and analyze its properties.

(a) [7 marks] Derive $\hat{\beta}_\lambda = \left(X^\top X + \lambda I\right)^{-1} X^\top y$ from the ridge objective $\min_\beta[\|y - X\beta\|^2 + \lambda\|\beta\|^2]$.

(b) [8 marks] Show that as $\lambda \to \infty$, $\hat{\beta}_\lambda \to 0$. Explain the geometric interpretation: what happens to the solution as regularization becomes infinitely strong?

Question 13. [10 marks] Explain Hadamard's well-posedness criteria and their connection to regularization.

(a) [5 marks] State the three criteria for a well-posed problem according to Hadamard. Give an example of a problem violating each criterion.

(b) [5 marks] How does Tikhonov regularization restore well-posedness to an ill-posed inverse problem? Which criterion does it primarily address?

Question 14. [15 marks] Application problem: comparing models for housing price prediction.

(a) [8 marks] You fit three models to predict housing prices from features:

- Model A (linear, 5 features): Train MSE = 2500, Test MSE = 2550
- Model B (polynomial, 20 features): Train MSE = 1200, Test MSE = 1300
- Model C (high-order, 100 features): Train MSE = 150, Test MSE = 5000

For each model, diagnose whether it's underfitting, well-fit, or overfitting. Justify using bias-variance reasoning.

(b) [7 marks] You decide to use ridge regression with Model C's feature set. Explain qualitatively how choosing λ affects the three components of MSE (bias, variance, irreducible error). What is the optimal λ criterion?

Question 15. [10 marks] Connect the bias-variance tradeoff to concepts from earlier weeks.

(a) [5 marks] In Week 2-3, we learned that larger hypothesis classes (higher VC dimension) can fit more patterns but require more data. Explain how this relates to bias and variance.

(b) [5 marks] How does the sample complexity bound $m = O\left(\frac{d}{\varepsilon^2}\right)$ for agnostic PAC learning relate to variance? What happens to variance as m increases for fixed d?

— End of Validation Sheet —